Khulna University of Engineering & Technology
B.Sc. Engineering 2nd Year 1st Term Examination, 2023
Department of Materials Science and Engineering

MSE 2101
(Crystallography)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Necessary: Tracing paper and Wulff Net

Section A
(Answer **ANY Three** questions from this section in **answer script A**)

1. CO1 (a) Write down the stacking sequence of HCP and FCC structures. (06)

CO2 (b) Describe the Nacl and Zinc blende structures with neat sketches. (14)

CO3 (c) Calculate the c/a ratio of HCP crystal and using this relationship, develop an equation for packing density of the HCP crystal. (15)

2. CO1 (a) Write down the steps for determination of crystal directions and planes. (10)

CO2 (b) Discuss the necessity of 4-coordinate system instead of conventional 3-coordinate system in case of hexagonal crystal system. (10)

CO3 (c) With the knowledge of Miller-Bravis indices, compute the indices of direction D, E, F from figure 2(c). (15)

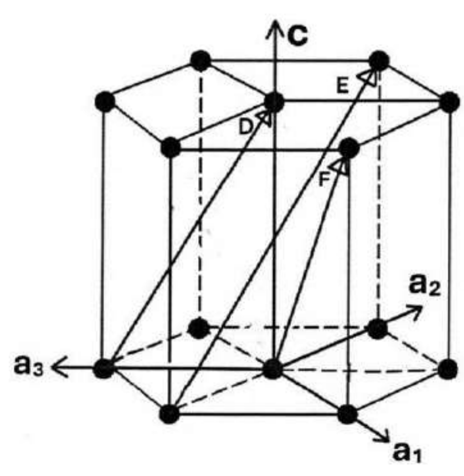

Figure 2(c)

3. CO1 (a) Define symmetry, symmetry operation and symmetry element. (06)

CO2 (b) With proper illustration explain the distinction between trigonal and hexagonal crystal structures though they possess similar lattice parameters of (15)

$$a = b \neq c, \alpha = \beta = 90^0, \gamma = 120^0.$$

CO3 (c) Calculate the planar density of most closed packed plane of FCC unit cell with identification of all such possible planes with proper illustration. (14)

4. CO1 (a) Find the all possible O-sites and T-sites in an FCC unit cell. (08)

CO2 (b) With neat sketches, discuss the structure of normal and inverse spinel structure. (15)

CO2 (c) Explain why it is not possible to have 5 or greater than 6-fold rotational symmetry for a 2-D crystal system. (12)

Section B

(Answer **ANY Three** questions from this section in **answer script B**)

5. CO1 (a) Define Bragg's angle, diffraction angle and order of reflection. (09)

CO2 (b) Explain how variable angles can be obtained in the Debye-Scherrer method. (13)

CO3 (c) For a certain BCC crystal, the (110) plane has a separation of $1.181 A^0$. These planes are indicated with x-rays of wavelength $1.540\ A^0$. Calculate how many order of Bragg's reflections can be observed in this case? (13)

6. CO1 (a) List the crystallographic informations those can be obtained from the Rietveld method. (05)

CO2 (b) Describe the working principle of neutron diffraction and explain how neutron diffraction differs from x-ray diffraction. (10+07)

CO3 (c) Derive the relationships between unit cell vectors of real lattice and reciprocal lattice for a monoclinic unit cell. (13)

7. CO1 (a) Define reciprocal lattice vector, and state the relationship between the d-spacing of real lattice and the d-spacing of reciprocal lattice. (03+04)

CO2 (b) Explain the reasons for peak broadening in any X-ray diffraction pattern. (10)

CO3 (c) Draw the plan view (with Miller indices) of a BCC crystal perpendicular to the z-axis. From this plan view, draw a two-dimensional reciprocal lattice and then extend it to a three-dimensional reciprocal unit cell. (18)

8. CO1 (a) Define primitive circle, great circle and small circle of a stereonet. (06)

CO2 (b) Explain Wigner-seitz cell and Brillouin zone with necessary figures. (08)

CO3 (c) i) Draw a stereographic projection of a cubic crystal on (001) plane using Tracing paper and Wulff Net, and then, (07+14)

(ii) Draw in the poles of $(111), and\ (1\bar{1}1)$ plane normal, label their traces and compute the angle between these two poles using Tracing paper and Wulff Net.

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Department of Materials Science and Engineering

MSE 2103
(Thermodynamics of Materials)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Necessary Table: Table 01:Properties of selected elements and Ellingham diagram.

Section A
(Answer **ANY THREE** questions from this section in **answer script A**)

1. CO1 (a) Explain the generic question addressed by thermodynamics "What happens?" by interpreting each labeled points in figure 1(a). (10)

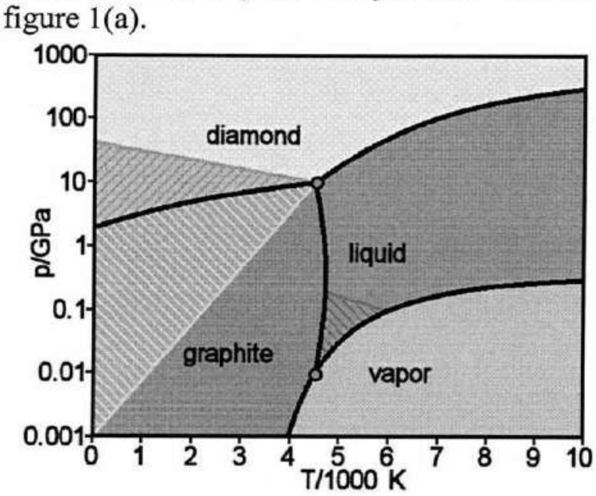

Figure 1(a)

CO2 (b) Classify the following thermodynamic systems (10)

i. A glass of hot milk.
ii. A can of Pepsi.
iii. A porous Styrofoam coffee cup.
iv. A Zirconia furnace tube.
v. A eutectic alloy of water turbine blade rotating at 100 rpm.

CO3 (c) Compare between process variables and state functions. The function z can be represented in terms of x and y by $z = 5x^4y^7 + 6x^6y^5$. Deduce whether z is a process variable. (15)

2. CO1 (a) Determine which of the following properties of a thermodynamic system are extensive properties and which are intensive: (i) Mass density, (ii) Molar density, (iii) Potential energy of a system in a gravitational field, (iv) Molar concentration on NaCl in salt solution, (v) Heat absorbed by the gas in a cylinder when it is compressed. (10)

CO3 (b) One of silver initially at 300K and one atmosphere pressure is taken through two separate processes. (i) An isobaric change in temperature to 500K, (ii) an isothermal compression to 1000 atm. (16)

Using table 01, compare the change in enthalpy of nickel for these two processes.

CO2 (c) "The first law of thermodynamics deals with the conservation of energy while the second law deals with the degradation of energy" – Explain. (09)

3. CO2 (a) Construct a schematic diagram with proper explanation to validate the third law of thermodynamics. (10)

CO1 (b) What type of thermodynamic variable is heat? Explain your answer. (10)

CO3 (c) One mole of monoatomic ideal gas, initially at 273K and 1atm, is contained in a chamber that permits programmed control of its state. Controlled quantities of heat and work are supplied in the system so that its pressure and volume change is given by the equation: V=22.4P (litre-atm). Assume the process is carried out reversibly. Evaluate the followings: (15)

(i) Heat transfer required to take the system to a final pressure of 0.5 atm. (ii) The final temperature of the gas.

4. CO2 (a) Using table 02, show that for an increase in temperature for a process in which the volume of the system is changed at constant entropy is $dT_s = -\frac{T\alpha}{C_V\beta} dV_s$. (12)

Table-02: Thermodynamic State Functions Expressed in Terms of the Independent Variables Temperature and Pressure

$V = V(T,P)$	$\mathrm{d}V = V\alpha \mathrm{d}T - V\beta\,\mathrm{d}P$
$S = S(T,P)$	$\mathrm{d}S = \frac{C_P}{T}\mathrm{d}T - V\alpha\,\mathrm{d}P$
$U = U(T,P)$	$\mathrm{d}U = (C_P - PV\alpha)\mathrm{d}T + V(p\beta - T\alpha)\mathrm{d}P$
$H = H(T,P)$	$\mathrm{d}H = C_P\mathrm{d}T + V(1 - T\alpha)\mathrm{d}P$
$F = F(T,P)$	$\mathrm{d}F = -(S + PV\alpha)\mathrm{d}T + PV\beta\,\mathrm{d}P$
$G = G(T,P)$	$\mathrm{d}G = -S\,\mathrm{d}T + V\,\mathrm{d}P$

CO3 (b) Deduce a constrained maximum for the function $z = xu + yv$, where $u = x + y + 20$ and $v = x - y - 4$ by (15)

(i) Eliminating dependent variables by direct substitution.

(ii) Eliminating the differentials of the dependent variables.

CO1 (c) Illustrate the structure of thermodynamics using a flowchart. (08)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. CO2 (a) Construct a unary phase diagram in (P,T) space to organize the characteristics of unary phase diagram in (P,T) space and develop relation between the Gibbs free energy and chemical potential. (12)

CO1 (b) Predict the Clasius-Clapeyron equation for a unary system in which sublimation occurs and deviation from ideal gas behavior is negligible. (13)

CO3 (c) Deuce the Van't Hoff isotherm using a general chemical reaction $lL + mM = rR + sS$. Evaluate this relation to the direction of the reaction. (10)

6. CO1 (a) Explain the graphical method to determine the partial molar properties from total properties. (10)

CO1 (b) Explain fugacity. Determine the Gibbs-Duhem equation for the activity coefficients in a binary system. (15)

CO3 (c) For a binary solution, partial molar enthalpy of component 2 is given by the equation (10)

$\Delta\overline{H}_2 = aX_1^2$. Compute the partial molar enthalpy for component 1 and the enthalpy of mixing for this solution.

7. CO1 (a) Define metastability. Explain metastability with respect to Gibbs free energy at constant temperature and pressure. (13)

CO2 (b) Define phase diagrams and show the allotropic transformation of iron. (10)

CO2 (c) A gas mixture at 1 atm total pressure and at the temperature 700°C has the following composition. (12)

Component	H_2	O_2	H_2O
Mole fraction	0.02	0.06	0.92

At 700°C, for the reaction: $2H_2 + O_2 = 2H_2O$, $\Delta G_0 = -440KJ$.

(i) Compute the direction of spontaneous change for this system

(ii) Develop the equilibrium composition for the system.

8. CO2 (a) Construct the equation of Gibb's phase rule. (10)

CO2 (b) Prepare a binary phase diagram of Pb-Sn metallic system and identify the phases present in that diagram. (10)

CO3 (c) Evaluate the following questions by consulting the Ellingham diagram: (15)

(i) Aluminium can be used to reduce titanium oxide at 1000°C whereas chromium cannot be used.

(ii) The CO_2 line is horizontal on the diagram but the CO line has a negative slope.

(iii) Carbon acts as an energetic reducing agent at high temperatures.

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Math 2127
(Vector Analysis and Linear Algebra)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A
(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) Define skew symmetric, skew hermitian and orthogonal matrix with example. (12)

(b) Show that every square matrix can be uniquely expressed as the sum of symmetric and skew-symmetric matrix. (12)

(c) Solve the system $2x + 3y + z = 9, x + 2y + 3z = 6, 3x + y + 2z = 8$ by the help on inverse. (11)

2. (a) Find the inverse of $A = \begin{bmatrix} 2 & 4 & 3 & 2 \\ 3 & 6 & 5 & 2 \\ 2 & 5 & 2 & -3 \\ 4 & 5 & 14 & 14 \end{bmatrix}$ by row transformation. (10)

(b) For what value of λ and μ the equations $x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu,$ have a no solution, unique solution and many solution. (13)

(c) Find all eigenvalues and all bases for the eigenspace corresponding to the largest eigenvalue of the matrix $\begin{bmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$. (12)

3. (a) Reduce the matrix $A = \begin{bmatrix} 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & 3 \\ 4 & 1 & -5 & 8 \\ 5 & -7 & 2 & -1 \end{bmatrix}$ to its echelon form and find its rank. (10)

(b) Find null A for $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$ and also find basis of null space and dimension. (12)

(c) Reduce $q = X'AX = X' \begin{bmatrix} 3 & 2 & 4 \\ 2 & -1 & -3 \\ 4 & -3 & 1 \end{bmatrix} X$ to the quadratic form where $X' = [x, y, z]$. (13)

4. (a) Determine the rank of the matrix $A = \begin{bmatrix} 0 & 2 & 2 & 1 \\ 3 & -2 & 0 & -1 \\ 1 & -2 & -3 & 2 \\ 0 & 1 & 2 & 1 \end{bmatrix}$. (13)

(b) Define vector space and subspace of a vector space with example. (12)

(c) Apply Cayley-Hamilton theorem to find the inverse of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$. (10)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Explain gradient, divergence and curl of a point function with their physical significance. (11)

   (b) Determine the rate of change of $\phi = xyz$ in the direction normal to the surface (12)

   $x^2y + y^2x + yz^2 = 3$ at the point $(1,1,1)$.

   (c) Evaluate the velocity and acceleration of a particle which moves along the curve $x = 2sin3t$, $y = 2cos3t$, $z = 8t$ at any time t>0. Hence, find the magnitude of the velocity and acceleration. (12)

6. (a) Find the constants a and b so that the surface $ax^2 - byz = (a+2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1,-1,2)$. (12)

   (b) Determine the constants a, b, c so that (12)

   $\vec{F} = (x + 2y + az)\hat{\imath} + (bx - 3y - z)\hat{\jmath} + (4x + cy + 2z)\hat{k}$

   is irrotational and hence find the corresponding scalar potential.

   (c) Verify Green's theorem in the plane for $\oint_c \{xy + y^2)dx + x^2dy\}$ where c is the closed curve of the region bounded by y=x and $y = x^2$. (11)

7. (a) Find the work done in moving a particle once around a circle C in the xy plane, if the circle has centre at the origin and radius 3 and the force field is given by (12)

   $\vec{F} = (2\mathrm{x} - \mathrm{y} + \mathrm{z})\hat{\imath} + (x + y - z^2)\hat{\jmath} + (3x - 2y + 4z)\hat{k}$.

   (b) Evaluate $\iint_s \vec{F}.\vec{n}ds$, where $\vec{F} = (x + y^2)\hat{\imath} - 2x\hat{\jmath} + 2yz\hat{k}$ and S is the surface of the plane $2x + y + 2z = 6$ in the first octant. (13)

   (c) Verify the divergence theorem for $\vec{A} = 4x\hat{\imath} - 2y^2\hat{\jmath} + z^2\hat{k}$ taken over the region bounded by $x^2 + y^2 = 4, z = 0$ *and* $z = 3$. (10)

8. (a) Define curvilinear coordinates and unit vectors in curvilinear system. (10)

   (b) Represent the vector $\vec{A} = 2y\hat{\imath} - z\hat{\jmath} + 3x\hat{k}$ in cylindrical coordinates. Thus determine (12)

   $A_p, A_\emptyset$ *and* A_Z.

   (c) Show that a cylindrical coordinate system is orthogonal. (13)

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Hum 2127
(Accounting and Economics)

Time: 3 hours | Full Marks: 210

N.B. (i) Answer **ANY Three** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.

Section A

(Answer **ANY Three** questions from this section in **answer script A**)

1. CO1 (a) Define accounting. How will you apply your accounting knowledge in your professional life? (10)

   CO1 (b) Distinguish between IFRS and GAAP. (07)

   CO2 (c) The following events occurred in "Messrs. MSE Enterprise" in April 2024: (18)

   April-1: Invested BDT 10,00,000 cash and furniture valued at BDT 3,00,000.
   April -7: Purchased goods for cash BDT 31,000.
   April -10: Goods were stolen from the shop valuing BDT 5,000.
   April -12: Paid an advertisement expense BDT 20,000 cash to the Daily Star .
   April -15: Cash deposited into bank BDT 70,000.
   April -17: Purchased furniture on account BDT 78,000.
   April -19: Cash withdrawal of BDT 17,500 by owner for personal use.
   April -22: Sold goods to 'Mansur Traders' and received a cheque of BDT 50,000.
   April -28: MSE Enterprise took a loan of BDT 3,00,000 from Janata Bank PLC, KUET Corporate Branch by signing a note.

   Required: Journalize the transactions for April 2024.

2. CO2 (a) Define the concept of trial balance. Discuss the objectives of trial balance. (10)

   CO2 (b) From the following ledger balances of "Salman Traders", prepare a trial balance as on 31 December, 2022. (25)

Particulars	BDT	Particulars	BDT
Capital	150,000	Debtors	30,000
Furniture	50,000	Salaries	5,000
Trademarks	30,000	Creditors	25,000
Bills Payable	25,000	Unearned Service Revenue	3,000
General Reserve	5,000	Bank Overdraft	2,000
Cash in hand (1-1-2022)	6,000	Goodwill	2,000
Inventory (1-1-2022)	40,000	Cash in Hand (31-12-2022)	8,000
Rent Expense	5,000	Insurance Premium	6,000
Provision for bad debt	3,000	Depreciation	2,000
Building	30,000	Inventory (31-12-2022)	55,000
Premises	15,000		

3. CO3 (a) Define Prime cost. (05)

CO3 (b) Distinguish between cost accounting and financial accounting. (10)

CO3 (c) The following data were taken from the records of "ExxonMobil Corporation" for the fiscal year ended on June 30,2020: (20)

Items	$	Items	$
Raw materials, 1/7/19	48,000	Office Utilities	8,650
Raw materials 30/6/20	39,600	Sales Revenue	534,000
Furnished Goods, 1/7/19	96,000	Plant manager's salary	4,200
Finished Goods 30/6/20	75,900	Factory Property taxes	58,000
Work in process 1/7/19	19,800	Factory repairs	9,600
Work in process, 30/6/20	18,600	Raw materials Purchase	96,400
Direct Labor	139,250		
Indirect Labor	24,460		
Factory Insurance	4,600		
Factory Machinary Depreciation	16,000		
Factory Utilities	27,600		

Required: (i) Prepare a cost Sheet

(ii) Prepare an income statement.

4. CO2 The following particulars are extracted from the book of "Mona Super Shop" relating to the year ended December 31, 2023: (35)

Mona Super Shop
Trial Balance
December 31, 2023

Account Titles	Debit(Tk.)	Credit(TK.)
Sales		18,00,000
Cash	70,000	
Accounts Payable		100,000
Accounts receivable	150,000	
Sales and purchase returns	50,000	25,000
Stock (1-1-2023)	100,000	
Capital and drawings	100,000	15,00,000
Carriage inward	40,000	
Administrative Expenses	250,000	
Wages	200,000	
Insurance	30,000	
Premises	12,00,000	
Equipment	200,000	
Allowance for Bad Debt		15,000
Purchases	950,000	
Selling and Distribution Expenses	100,000	
Total=	34,40,000	34,40,000

Additional Information:

(i) Closing stock is TK. 80,000.
(ii) TK. 20,000 accrued for administrative expenses during 2023.
(iii) Create an allowance for bad debt @10% on accounts receivable.
(iv) Equipment is to be depreciated @20%

Required: Prepare a

(i) Statement of comprehensive income
(ii) Statement of owner's equity.
(iii) Statement of Financial position.

Section B

(Answer **ANY Three** questions from this section in **answer script B**)

5.	CO1	(a)	Define Macroeconomics. Discuss the scope of economics according to the traditional approach.	(15)
	CO1	(b)	Distinguish between the deductive method and the inductive method of economic analysis.	(10)
	CO1	(c)	What is meant by demand law? Why does the demand curve shift?	(10)
6.	CO1	(a)	Explain the equilibrium of a market.	(15)
	CO1	(b)	Fahim advertises to sell cookies for 40 TK a dozen. He sells 50 dozen and decides that he can charge more. He raises the price to 60TK a dozen and sells 40 dozen. What is the price elasticity of demand? What kind of elasticity of demand is it?	(05)
	CO3	(c)	"Higher savings lead to a higher level of investment"-Interpret.	(15)
7.	CO3	(a)	Explain the different kinds of laws of variable proportion?	(15)
	CO3	(b)	Define shut down position. Explain the shut down position of a firm under perfect competition with figure.	(15)
	CO3	(c)	Why is a firm in a monopoly market a price maker? Discuss.	(05)
8.	CO3	(a)	How does the cost-push inflation occur in an economy? Illustrate.	(10)
	CO3	(b)	What are the remedies for fighting inflation with fiscal policy? Explain.	(15)
	CO3	(c)	Distinguish between GNP and GDP.	(10)

Khulna University of Engineering & Technology

B.Sc. Engineering 2nd Year 1st Term Examination, 2023

Department of Materials Science and Engineering

ME 2127

(Solid Mechanics)

Time: 3 hours Full Marks: 210

N.B. (i) Answer **ANY THREE** questions from each section in separate scripts.
(ii) Figure in the right margin indicates full marks.
(iii) Necessary Table: Table B-2: Properties of Wide-Flange Section.
(iv) Instruction for the candidate (if any): Assume reasonable data if any missing.

Section A

(Answer **ANY THREE** questions from this section in **answer script A**)

1. (a) State Hook's law for axial deformation and also mention its limitation (06)

(b) Show that the tangential stress in a thin walled cylindrical shell of diameter 'D' and wall thickness 't' subjected to internal pressure 'P' is given by $\sigma_t = {PD}/{2t}$. (12)

(c) Referring to the figure 1(c), compute the maximum force P that can be applied by the machine operator if the shearing stress in the pin at B and the axial stress in the control rod at C are limited to 30 MPa snd 35 MPa respectively. The diameters are 7 mm for the pin and 35 mm for the control rod. Assume single shear for the pin at B. (17)

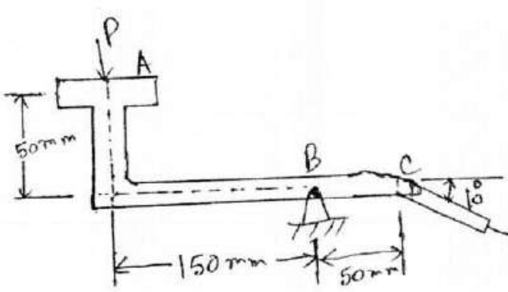

Figure 1(c)

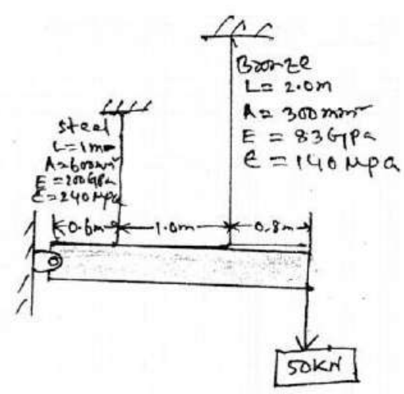

Figure 2(b)

2. (a) Explain the following terms: (i) Shear stress, (ii) Bearing stress, (iii) Normal stress, and (iv) Yield strength. (10)

(b) A horizontal bar of negligible mass, hinged at A shown in figure 2(b) and assumed rigid. Compute the stress in each rod. (13)

(c) The composite bar shown in figure 2(c) s firmly attached to unyielding supports. Assuming the structure is stress free before the axial load P_1 and P_2 are applied. Calculate the stress in each material if P_1=150 KN and P_2=90 KN. (12)

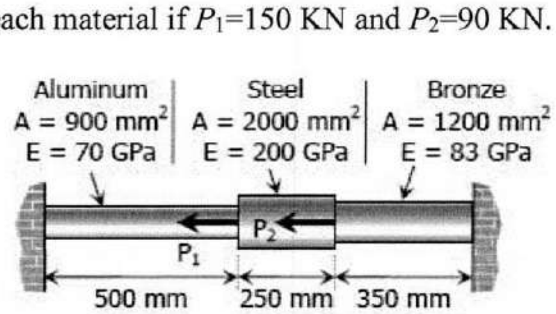

Figure 2(c)

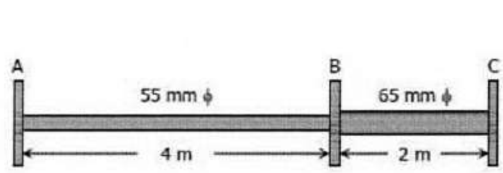

Figure 3(a)

3. (a) The steel shaft shown in figure 3(a) rotates at 4 Hz with 35 KW taken off at *A*. 20 KW removed at *B* and 55KW applied at *C*. Using G=83 GPa, find the maximum shearing stress and the angle of rotation of gear *A* relative to gear *C*. (18)

(b) Determine the maximum shearing stress and elongation in a helical steel spring composed of 20 turns of 20 mm diameter wire on a mean radius of 90 mm. When the spring is supporting a oad of 1.5 KN. (17)

4. (a) Write shear and moment equations for the beam loaded as shown in figure 4(a) and sketch the shear and moment diagrams. (18)

(b) Determine the minimum height *h* of the beam shown in figure 4(b) if the flexural stress in not to exceed 20 MPa. (17)

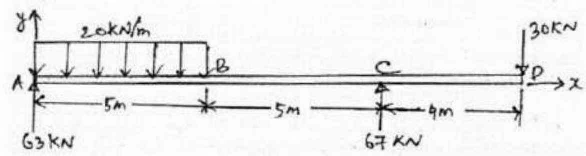

Figure 4(a)

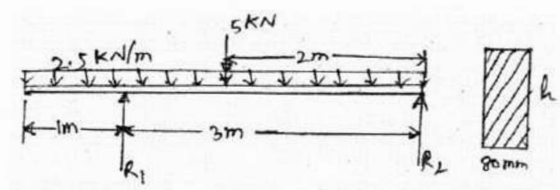

Figure 4(b)

Section B

(Answer **ANY THREE** questions from this section in **answer script B**)

5. (a) Derive Eular formula for critical load of a hinged end column. (17)

(b) Using AISC column specifications, determine the safe axial loads on a W360 × 122 section used as a column under the following conditions: (i) hinged ends and a length of 9m; (ii) built in ends and an unsupported length of 10 m; (iii) built-in ends and a length of 10m braced at the midpoint. Use σ_{yp}=380MPa and *E*=200 GPa. (Use Table B-2: Properties of Wide-Flange Section.) (18)

6. (a) Determine the centroid of the area as shown in figure 6(a) by direct integration. (17)

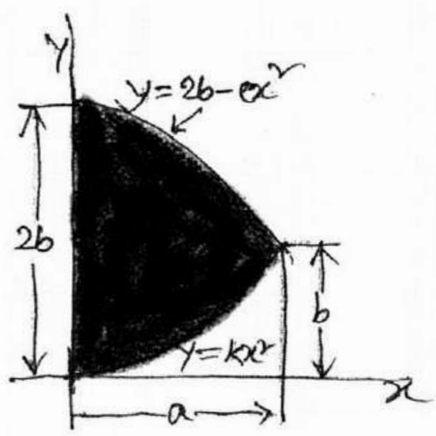

Figure 6(a)

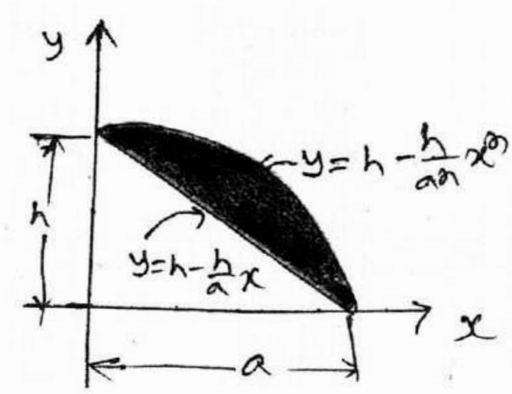

Figure 6(b)

(b) Determine the centroid $(\overline{x}, \overline{y})$ of the shaded area as shown in figure 6(b). (18)

7. (a) Determine the force in each member of the truss as shown in figure 7(a) and state if the members are in tension or compression. (17)

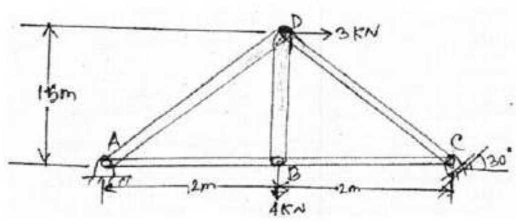

Figure 7(a)

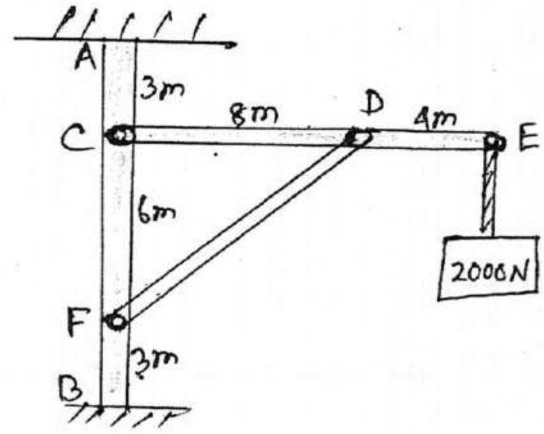

Figure 7(b)

(b) Each member of the frame as shown in figure 7(b) weights 50N per m. Compute the horizontal and vertical components of the pin pressure at C, D and F. (18)

8. (a) Determine by direct integration the moment of inertia of the shaded area as shown in figure 8(a) with respect to each of the coordinate axes. (17)

(b) Determine product of inertia of the right triangle as shown in figure 8(b), (i) with respect to the x and y axis. (ii) with respect to centroidal axis parallel to the x and y axis. (18)

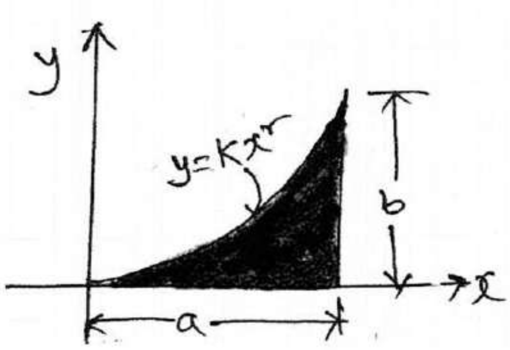

Figure 8(a)

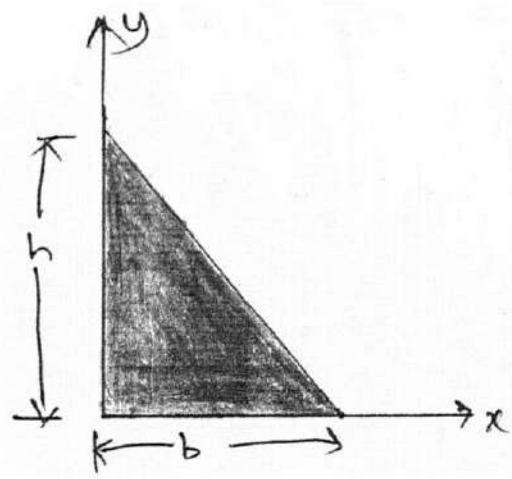

Figure 8(b)